

Review Problems

Problem 1. Given parametric equations $x = 3t^2$, $y = 4t^3$, $t \neq 0$. Without eliminating the parameter,

- a) Find dy/dx .
- b) Find d^2y/dx^2 .

Problem 2. Let $\mathbf{a} = 5\mathbf{i} - 2\mathbf{j} + 5\mathbf{k}$, $\mathbf{b} = 3\mathbf{i} - \mathbf{k}$, and $\mathbf{c} = \langle 1, 6, 0 \rangle$, Find each of the following:

- a) The angle between $\mathbf{a}$ and $\mathbf{b}$.
- b) $\mathbf{b} \times \mathbf{c}$.
- c) $\mathbf{c} \cdot \mathbf{c} - |\mathbf{c}|^2$.
- d) The projection of $\mathbf{c}$ on $\mathbf{a}$, $Pr_{\mathbf{a}}\mathbf{c}$.

Problem 3.

- a) What is the direction of the line $x = 2 - 3t$, $y = 3t$, $z = 2 - t$?
- b) Find the parametric equations for the line through $(6, 1, -3)$ and $(-2, 1, 3)$
- c) Find an equation for the plane containing the lines $x = 3t$, $y = 1 + t$, $z = 4 + t$ and $x = 3 - 2s$, $y = 2$, $z = 5 + 4s$.

Problem 4. Two vectors $\mathbf{u} = 2\mathbf{i} + z\mathbf{j} + 2\mathbf{k}$ and $\mathbf{v} = \langle 2, 4, -3 \rangle$ are perpendicular, find z .

Problem 5. Consider the curve $\mathbf{r}(t) = (\cos t + t \sin t)\mathbf{i} + (\sin t - t \cos t)\mathbf{j} + t\mathbf{k}$, find each of the following:

- a) $\mathbf{r}'(t)$ and $|\mathbf{r}'(t)|$.
- b) The unit tangent vector $\mathbf{T}(\pi/2)$ to the curve.
- c) The length of the curve from $t = 0$ to $t = 2\pi$.

Problem 6.

- a) Find the center and radius of the sphere with equation $x^2 + y^2 + z^2 + 2x - 8z + 1 = 0$.
- b) Change $(4, 4\pi/3, -8)$ from cylindrical to Cartesian and spherical coordinates.
- c) Change the Cartesian equation $x^2 + y^2 + 4z^2 = 10$ to spherical and cylindrical coordinate equation.

Problem 7. (15 points) Given a function $f(x, y) = y \cos(x^2 - y^2)$

- a) Find its two first partial derivatives $f_x(x, y)$ and $f_y(x, y)$.
- b) Find its four second partial derivatives $f_{xx}(x, y)$, $f_{yy}(x, y)$, $f_{xy}(x, y)$, $f_{yx}(x, y)$.

Problem 8. Let $z = \frac{1}{2}x^2 + y^2$, sketch the level curves of $z = k$ for $k = 0, 1, 4, 9$ in the plane. (Remember to label the curves and the axes.)

Problem 9.

- a) Given $f(x, y) = x - 4xy + 2y^2$, $\mathbf{p} = (-1, 2)$ and $\mathbf{a} = \sqrt{3}\mathbf{i} + \mathbf{j}$, find the directional derivative of f at the point $\mathbf{p}$ in the direction of $\mathbf{a}$.
- b) Given a surface determined by $x^2 + y^2 - z^2 = 0$, find the equation of the tangent plane to this surface at the point $(2, 1, 1)$.

Problem 10. Let $w = \ln(2x + 3y)$, $x = te^s$, $y = e^{st}$, find $\frac{\partial w}{\partial t}$ and $\frac{\partial w}{\partial s}$ using chain rule and express the final answer in term of s and t .

Problem 11. Given a function $f(x, y) = e^{-(x^2+y^2-4y)}$,

- a) Find all critical points of f .
 b) Determine whether each such a point gives a local maximum or minimum, or whether it is a saddle point using the **second partials test**.

Problem 12. (15 points) Find the minimum of $f(x, y, z) = 4x - 2y + 3z$ subject to the constraint $2x^2 + 3y - 2z = 4$ using **Lagrange's method**.

Problem 13.

a) Evaluate $\int_{-2}^2 \int_{-\sqrt{4-y^2}}^{\sqrt{4-y^2}} 2xy^2 \, dx \, dy$.

b) Evaluate $\int_1^4 \int_3^x \int_0^{\sqrt{3y}} 2xyz \, dz \, dy \, dx$.

Problem 14.

a) Evaluate $\iint_S \frac{1}{x^2 + y^2} \, dA$ where S is the region between the circles $x^2 + y^2 = 4$ and $x^2 + y^2 = 9$ using polar coordinates..

b) Find the mass m and the center $(\bar{x}, \bar{y})$ of the lamina bounded by the curves $y = x$ and $y = 3x - x^2$ with the density $\delta(x, y) = x + y$.

Problem 15. Let S be the solid bounded below by the paraboloid $z = \sqrt{x^2 + y^2}$ and above by the sphere $x^2 + y^2 + z^2 = 4$. The iterated triple integral in Cartesian coordinates for the volume of S is

$$V(S) = \int_{-\sqrt{2}}^{\sqrt{2}} \int_{-\sqrt{2-x^2}}^{\sqrt{2-x^2}} \int_{\sqrt{x^2+y^2}}^{\sqrt{4-x^2-y^2}} dz \, dy \, dx$$

- a) Convert the triple integral for the volume of S to cylindrical coordinates.
 b) Convert the triple integral for the volume of S to spherical coordinates.
 c) Evaluate one of the three triple integrals to find the volume of S .

Problem 16. Let $\mathbf{F}(x, y, z) = 2xyz\mathbf{i} - 3y^2\mathbf{j} + 2y^2z\mathbf{k}$. Find

- a) $\text{div}\mathbf{F}$.
 b) $\text{curl}\mathbf{F}$.
 c) $\nabla(\text{div}\mathbf{F})$.
 d) Is $\mathbf{F}$ a conservative vector field?

Problem 17. Find the work done by the force field $\mathbf{F}$ in moving a particle along the curve C where $\mathbf{F}(x, y) = y\mathbf{i} + z\mathbf{j} + x\mathbf{k}$ and C is the curve $x = t$, $y = t^2$, $z = t^3$, $1 \leq t \leq 2$.